

Touchscreen Control for Wheelchair

SACHIDANANDA SAHU

A wheelchair is a chair with wheels, designed to be a replacement for walking. The device comes in variations where it is propelled by motors or by the seated occupant turning the rear wheels by hand. Wheelchairs are used by people for whom walking is difficult or impossible due to illness, injury or disability.

Here we describe a microcontroller-based wheelchair the speed and direction of which can be controlled

from a touchscreen. The wheelchair moves by means of a geared motor. Fig. 1 shows the block diagram of the touchscreen-controlled wheelchair. The system includes a resistive touchscreen, microcontroller and motor driver circuit.

The touchscreen

The touchscreen is an electronic visual display that can detect the presence and location of a touch within the display area. It is sensitive to the touch of a human finger, hand and passive

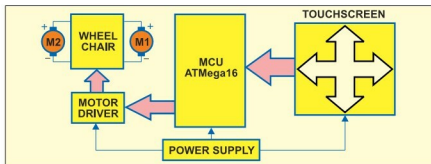


Fig. 1: Block diagram for the touchscreen control for wheelchair

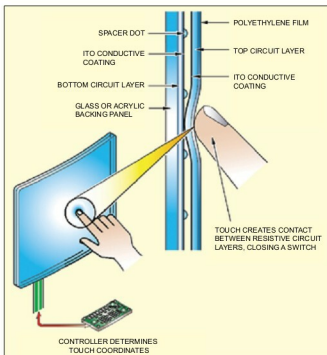


Fig. 2: Typical touchscreen layers (Courtesy: www.tci.de)

layer made of glass. Both the layers are coated with a conducting compound of indium-tin oxide (ITO) and then spaced with spacers. When the monitor is operational, an electric current flows between the two layers.

When a touch is made, the flexible screen presses down to touch the bottom layer. A change in electrical current is hence detected and the coordinates of the touch point are calculated by the controller and parsed into readable signals for the operating system to react accordingly.

The 4-wire resistive touchscreen uses both the layers to calculate the axes information of the touch. Touch measurement is a two-step process. The x-coordinate of the touch point is calculated by creating a voltage gradient on the flexible layer. The y-coordinate is determined by creating a voltage gradient along the bottom layer.

Capacitive touchscreen. Capacitive touchscreen is the most popular and durable touchscreen technology used all over the world. It consists of a glass panel coated with indium-tin oxide—a capacitive (conductive) material. Capacitive systems transmit almost 90 per cent of light from the monitor. In surface-capacitive screens, only one side of the insulator is coated with a conducting layer.

When the monitor is operational, a uniform electrostatic field is formed over the conductive layer. Whenever the user touches the screen with a finger, conduction of electric charges over the uncoated layer results in the formation of a dynamic capacitor. The computer or the controller then detects the position of touch by measuring

objects like stylus (refer Fig. 2). It is a two-dimensional sensing device which is made of two sheets of material separated by small spacers.

There are three main touchscreen technologies: resistive, capacitive and surface acoustic wave.

Resistive touchscreen. The resistive touchscreen consists of a flexible top layer made of polyethylene and a rigid bottom